PVsyst - Simulation report

Grid-Connected System

Project: 100 MW Solar PV Grid Connected

Variant: New simulation variant

Sheds on ground due to wind turbines

System power: 108.7 MWp

Dholera - India

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Project summary

Geographical Site	**Situation**		**Project settings**	
Dholera	Latitude	22.2493 °(N)	Albedo	0.20
India	Longitude	72.1942 °(E)		
	Altitude	14 m		
	Time zone	UTC+5.5		

Weather data
Dholera
Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System
Simulation for year no 10

Sheds on ground due to wind turbines

Orientation #1
Fixed plane
Tilt/Azimuth 21 / 0 °

Near Shadings
Linear shadings : Fast (table)

User's needs
Unlimited load (grid)

System information

PV Array		**Inverters**	
Nb. of modules	190706 units	Nb. of units	50 units
Pnom total	108.7 MWp	Total power	100000 kWac
		Pnom ratio	1.09

Results summary

Simulation step	Hourly				
Produced Energy	168067641 kWh/year	Specific production	1546 kWh/kWp/year	Perf. Ratio PR	76.73 %

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General parameters

Grid-Connected System — **Sheds on ground due to wind turbines**

Simulation step
Hourly

Orientation #1

Fixed plane		Sheds configuration		Sizes	
Tilt/Azimuth	21 / 0 °	Nb. of sheds	837 units	Sheds spacing	8.00 m
		Identical arrays		Sensitive width	2.29 m
		Shading limit angle		GCR Shading	28.9 %
		Limit profile angle	8.1 °	Top inactive band	0.02 m
				Bottom inactive band	0.02 m

Models used		Horizon	Near Shadings
Transposition	Perez	Free Horizon	Linear shadings : Fast (table)
Diffuse	Perez, Meteonorm		
Circumsolar	separate		

User's needs
Unlimited load (grid)

PV Array Characteristics

PV module		Inverter	
Manufacturer	Generic	Manufacturer	Generic
Model	ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX	Model	Solar Inverter DelCEN 1000x2
(Custom parameters definition)		(Original PVsyst database)	
ReNew_M10_144_570_TOPCon_Bifacial_RPS2MH72BDXXX.PAN		Unit Nom. Power	2000 kWac
Unit Nom. Power	570 Wp	Number of inverters	50 units
Number of PV modules	190706 units	Total power	100000 kWac
Nominal (STC)	108.7 MWp	Operating voltage	610-930 V
Modules	11218 units x 17 In series	Max. power (=>40°C)	2200 kWac
At operating cond. (50°C)		Pnom ratio (DC:AC)	1.09
Pmpp	100.0 MWp		
U mpp	688 V		
I mpp	145375 A		

Total PV power		Total inverter power	
Nominal (STC)	108702 kWp	Total power	100000 kWac
Total	190706 modules	Max. power	110000 kWac
Module area	492642 m²	Number of inverters	50 units
Cell area	454820 m²	Pnom ratio	1.09

Array losses

Array Soiling Losses		Thermal Loss factor		DC wiring losses	
Loss Fraction	2.0 %	Module temperature according to irradiance		Global array res.	0.078 mΩ
		Uc (const)	29.0 W/m²K	Loss Fraction	1.50 % at STC
		Uv (wind)	0.0 W/m²K/m/s		

LID - Light Induced Degradation		Module Quality Loss		Module mismatch losses	
Loss Fraction	2.00 %	Loss Fraction	-0.22 %	Loss Fraction	2.00 % at MPP

Array losses

Strings Mismatch loss
Loss Fraction 0.15 %

Module average degradation
Year no 10
Loss factor 0.4 %/year
Imp / Vmp contributions 80% / 20%

Mismatch due to degradation
Imp RMS dispersion 0.4 %/year
Vmp RMS dispersion 0.4 %/year

IAM loss factor
Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000

Spectral correction
FirstSolar model
Precipitable water estimated from relative humidity

Coefficient Set	C0	C1	C2	C3	C4	C5
Monocrystalline Si	0.85914	-0.02088	-0.0058853	0.12029	0.026814	-0.001781

System losses

Unavailability of the system
Time fraction 1.0 %
3.7 days,
3 periods

Auxiliary losses

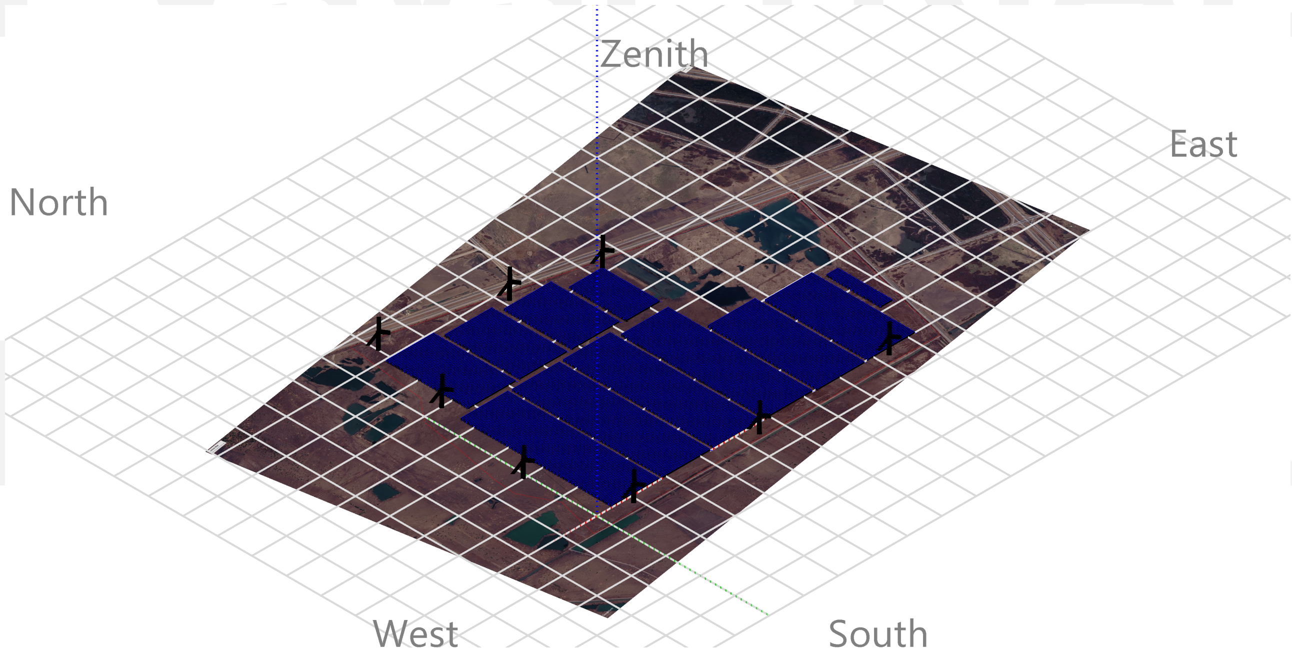
AC wiring losses

Inv. output line up to injection point
Inverter voltage 400 Vac tri
Loss Fraction 0.00 % at STC

Inverter: Solar Inverter DelCEN 1000x2
Wire section (50 Inv.) Copper 50 x 3 x 4000 mm²
Average wires length 0 m

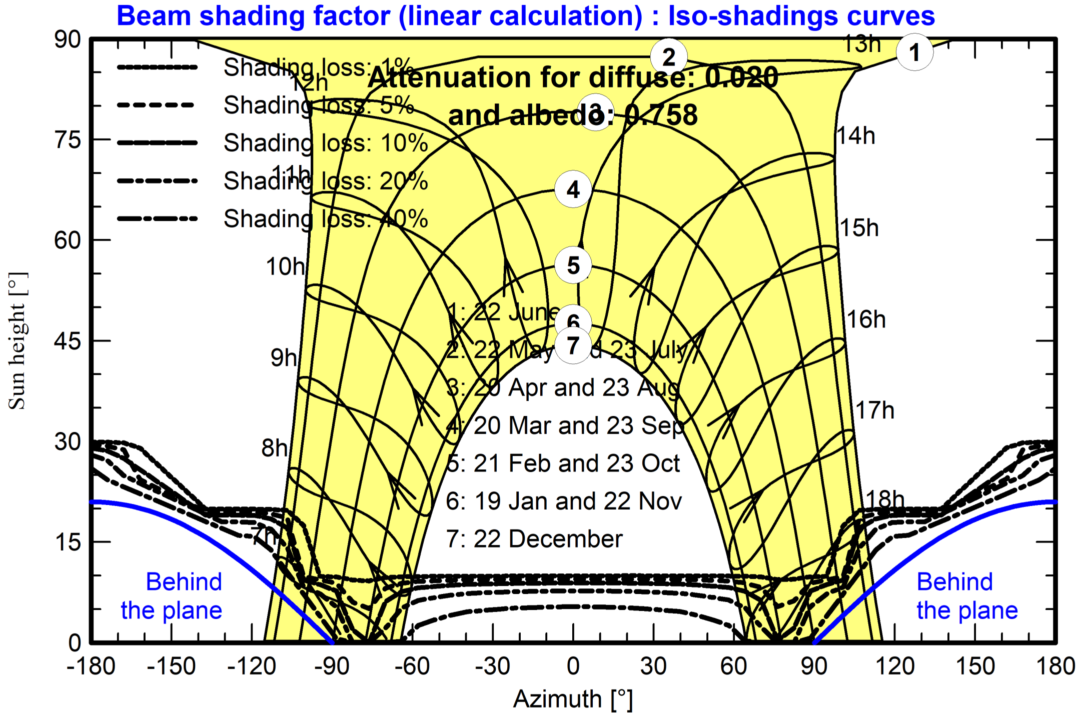
Near shadings parameter

Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1 - Fixed plane, Tilts/azimuths: 21°/ 0°



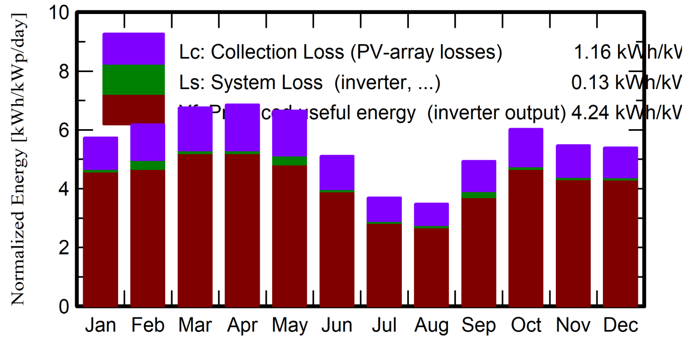
Main results

Simulation step
Hourly

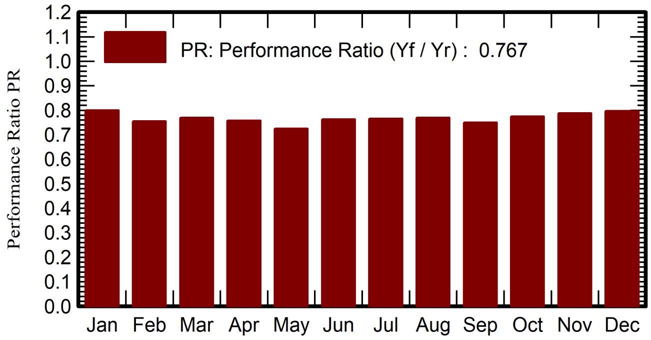
System Production

Produced Energy	168067641 kWh/year	Specific production	1546 kWh/kWp/year
		Perf. Ratio PR	76.73 %

Normalized productions (per installed kWp)



Performance Ratio PR



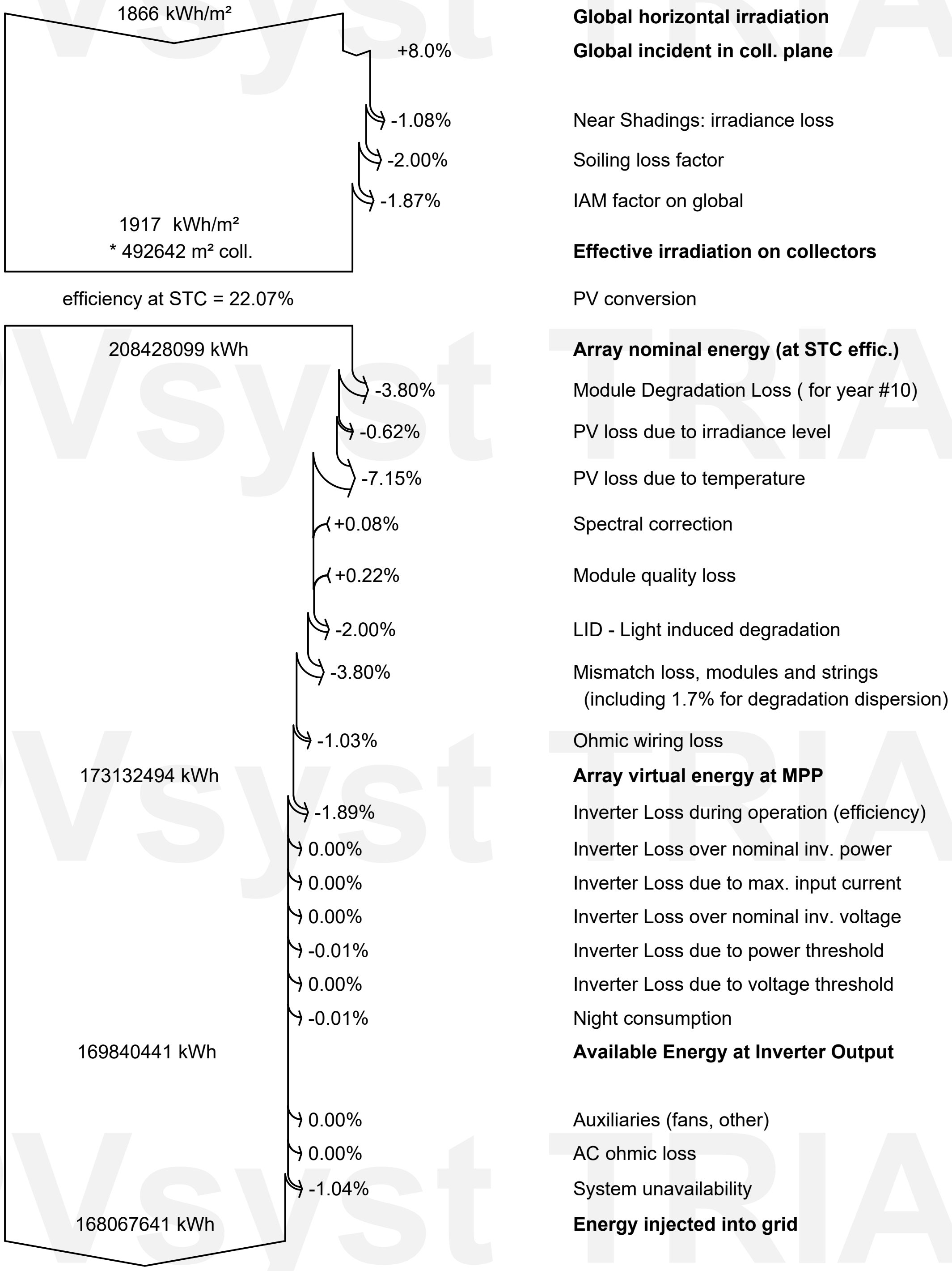
Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m²	kWh/m²	°C	kWh/m²	kWh/m²	kWh	kWh	ratio
January	138.6	41.00	21.43	177.7	169.8	15732384	15444756	0.800
February	145.4	49.86	24.34	173.2	165.6	15123585	14201130	0.754
March	191.6	63.86	28.71	209.7	200.3	17846649	17535598	0.769
April	204.6	75.30	32.33	205.9	196.1	17274789	16969122	0.758
May	218.8	87.88	33.65	206.1	195.8	17254822	16244036	0.725
June	165.5	98.20	32.10	153.4	144.9	12991334	12728425	0.764
July	122.2	81.18	29.54	114.6	107.8	9782775	9539998	0.766
August	111.9	78.39	28.50	108.1	101.7	9274825	9040469	0.769
September	142.9	82.05	29.27	148.1	140.7	12745701	12068525	0.750
October	163.3	63.25	30.09	186.9	178.2	16012894	15730895	0.774
November	132.7	55.29	26.74	164.3	156.4	14319353	14063722	0.788
December	128.0	41.60	22.80	167.3	159.6	14773157	14500965	0.797
Year	1865.6	817.86	28.30	2015.1	1916.8	173132268	168067641	0.767

Legends

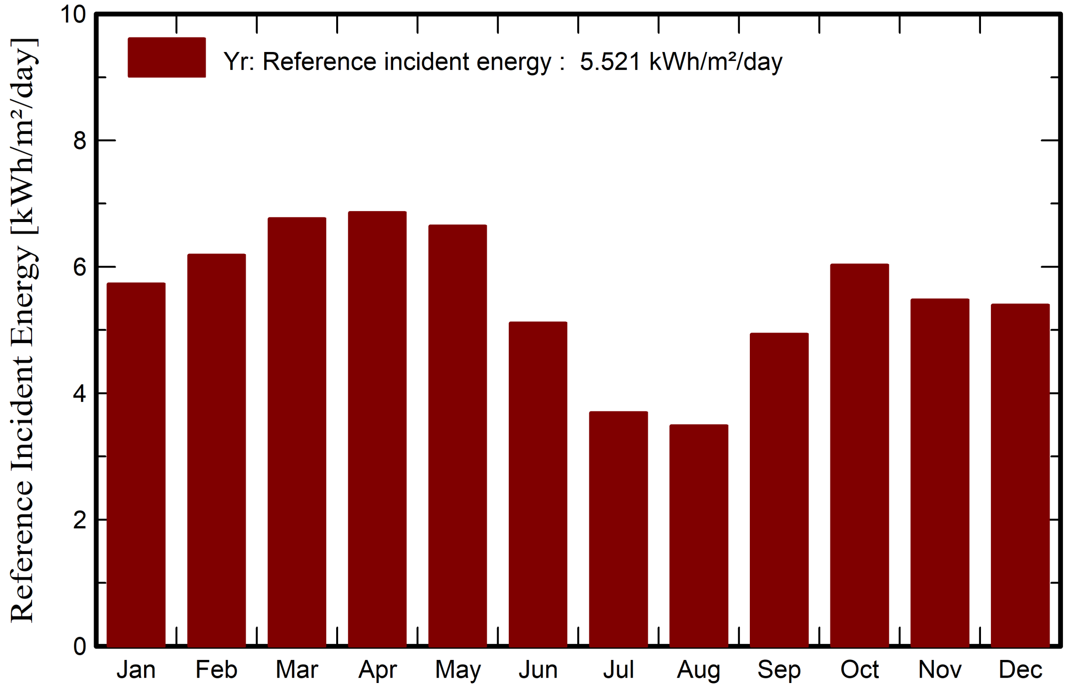
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		

Loss diagram

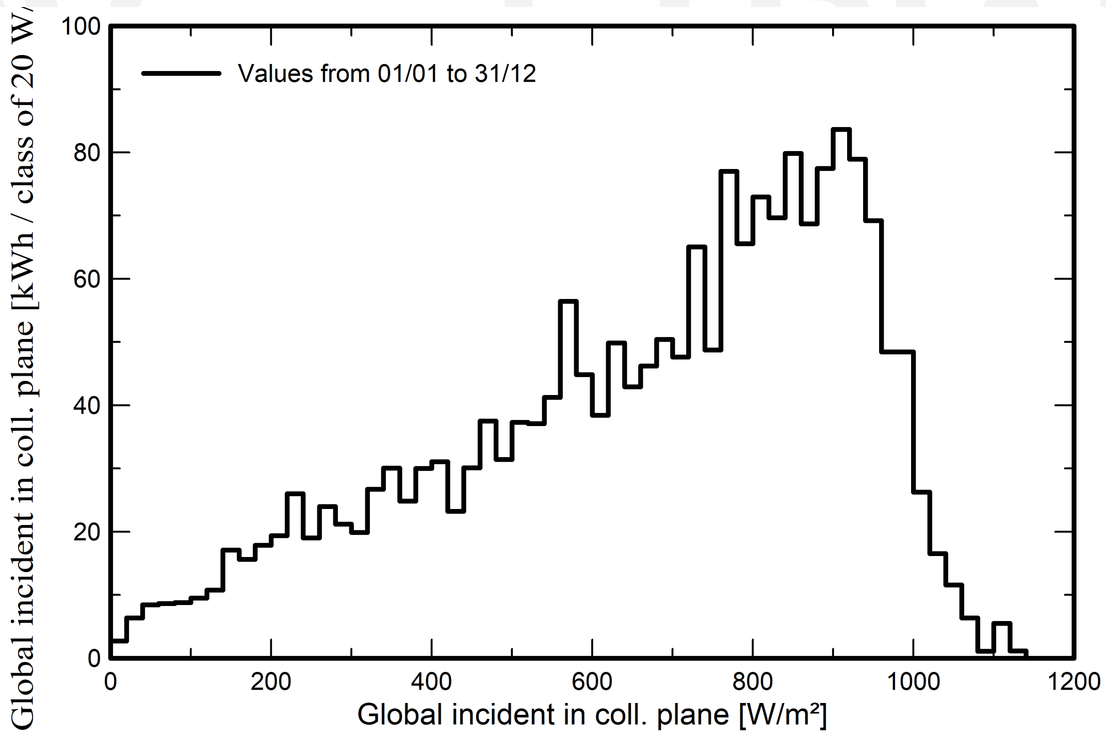


Predef. graphs

Reference Incident Energy in Collector Plane

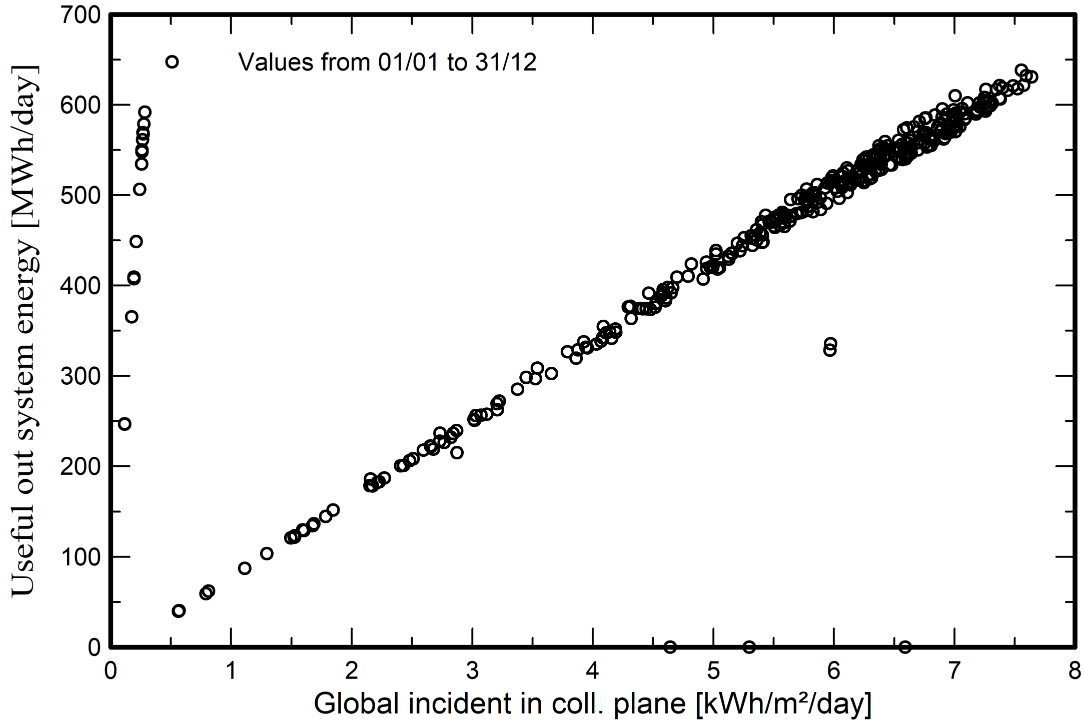


Incident Irradiation Distribution

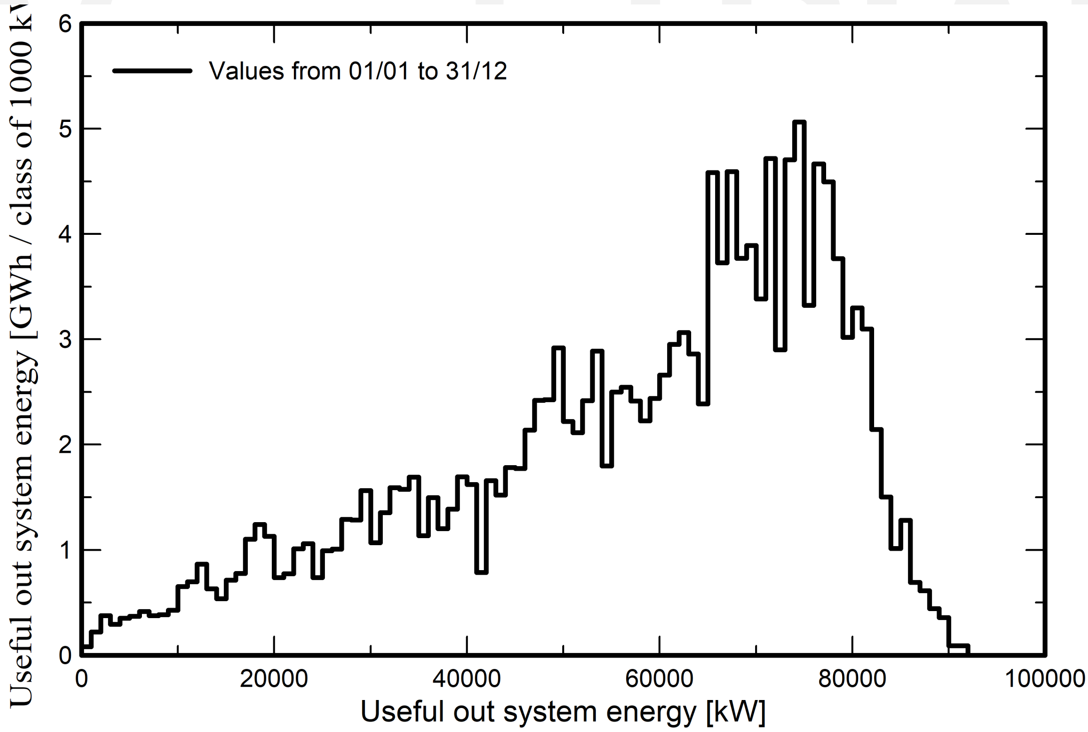


Predef. graphs

Daily Input/Output diagram



System Output Power Distribution



P50 - P90 evaluation

Weather data

Source	Meteonorm 9.0 dll, Sat=100%
Kind	Not defined
Year-to-year variability(Variance)	0.0 %

Specified Deviation

Global variability (weather data + system)

Variability (Quadratic sum)	1.8 %

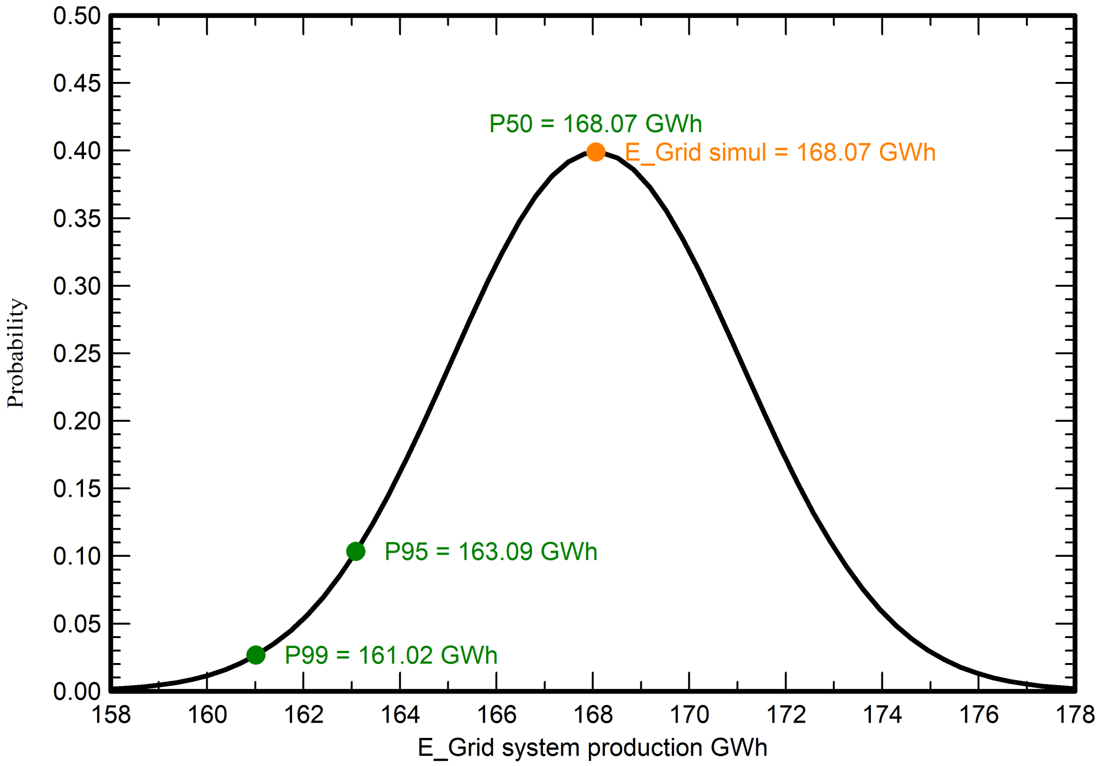
Simulation and parameters uncertainties

PV module modelling/parameters	1.0 %
Inverter efficiency uncertainty	0.5 %
Soiling and mismatch uncertainties	1.0 %
Degradation uncertainty	1.0 %

Annual production probability

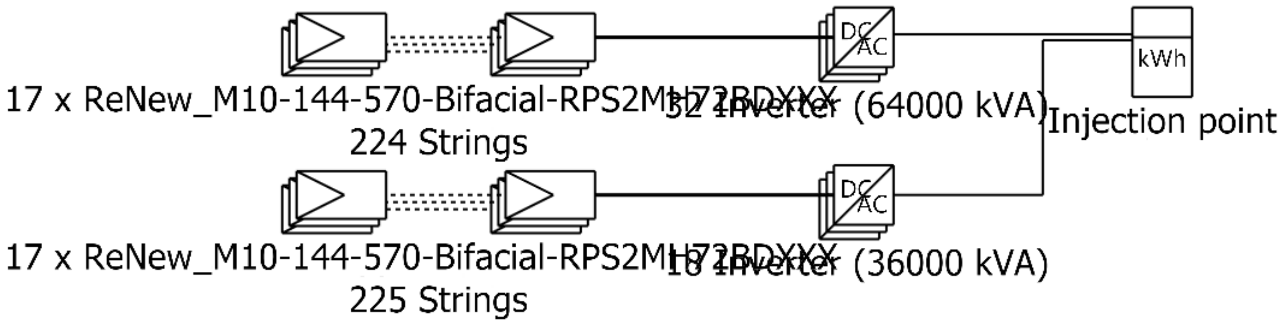
Variability	3.03 GWh
P50	168.07 GWh
P99	161.02 GWh
P95	163.09 GWh

Probability distribution



PVsyst V8.1.3
VC0, Simulation date:
31/05/26 18:14
with V8.1.3

Single-line diagram



PV module	ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX
Inverter	Solar Inverter DelCEN 1000x2
String	17 x ReNew_M10-144-570-Bifacial-RPS2MH72BDXXX

100 MW Solar PV Grid Connected	
VC0 : New simulation variant	31/05/26